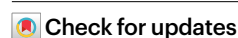


Degrowth as a plausible pathway for food systems transformation

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Food systems require urgent transformation towards social and ecological sustainability. Degrowth posits a radical pathway of transformation to reduce ecological impacts while increasing well-being and reducing inequality. Here we highlight that degrowth and food systems—albeit both linked to transformation—are not well integrated. We conduct a conceptual exploration of the potential alignment between key food systems and degrowth transformation measures, arguing for complementary and reciprocal perspectives to theorize and enact transformation. Finally, we offer concrete practical actions to integrate degrowth and food systems, thereby widening the narrative and analytical lens of social–ecological transformation.

Degrowth posits a planned equitable downscaling of excess production and consumption among high-income countries and overconsuming individuals to increase well-being and reduce ecological impact and inequality¹. Degrowth proponents share a common critique of dominant growth-centric economic systems, advancing core propositions that perpetual economic growth is unsustainable and that reduced inequality and improved human well-being are possible without global aggregate economic growth^{2,3}. While these overall goals and views are shared by the degrowth community, degrowth research is itself broad, including, for example, ecological economics, notions of voluntary simplicity and critiques of the western development paradigm^{4,5}.

A separate body of work has argued for the urgent need to transform food systems^{6–8}. Central to many transformation proposals is a redesign of consumption, production and waste to deliver healthy diets from sustainable, just food systems⁹. However, the nature of this transformation is deeply contested, with powerful actors working to maintain the status quo or shape future pathways in their interest^{10–12}. Here we do not argue for any one conceptualization of transformation but for the need to widen the lens through which we imagine, analyse and enact transformation. We also argue that the integration of degrowth and food systems is a potentially fruitful frontier to do so.

Alternative conceptualizations of food systems transformation have included considerations for sustainable agri-food systems of a

post-growth world and calls for academics to take up the challenge of reframing the political debate for food systems transformation^{13,14}. We depart from and build on this work by taking a practical approach to integrate degrowth and food systems perspectives to build deeper, more diverse pathways towards desirable social–ecological transformation.

Limited integration of degrowth and food systems transformation

While expanding, academic literature on degrowth and food systems remains nascent. Explicit literature on ‘degrowth’ and ‘food systems’ or ‘degrowth’ and ‘food systems transformation’ is less than 1% of their respective larger literature bases (see Supplementary Information for the methods). While we are cognizant that much research probably considers similar concepts but does not name it as such, this small volume of published academic literature indicates clear underexplored terrain.

For a wider view on the inclusion of degrowth, we conducted content analysis of 52 global food systems policy documents spanning 2016–2023 (see Supplementary Information for the full list and the selection of the time period). The list of publications to extract for analysis was generated using the search strategies and results of previous work on analysing global food systems reports^{15,16}. The method of content analysis we employed follows that of recent literature assessing

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Goals and objectives of transformation



Fig. 1 | Alignment between goals and objectives of degrowth and food systems transformation. There is in general clear potential alignment between food systems transformation and common degrowth measures (sourced from ref. 19). Connections between degrowth measures (left) and food systems (right) are coloured blue for potentially aligned, grey for a clear interaction but an unclear alignment and red for potentially unaligned. One potential connection is sufficiency (box 10 on the left), which is both a degrowth topic and a standalone concept about sustainable production and consumption and the sufficing of human activity in relation to the more-than-human world. There is clear alignment within food systems with the former (for example, reducing animal-source foods and reducing food loss and waste), whereas the latter has largely been seen through a health lens of sufficiency of dietary demand rather than a broader ecological

perspective. Accounting for true cost (box 15 on the right) is potentially aligned with beyond-GDP indicators and ecological taxes. However, this depends on its implementation. Degrowth would be at odds with efforts to further marketize, financialize and commodify the social and ecological realms to provide ‘transparent’ and ‘true’ costs borne by ‘better-informed’ citizens. Indeed, degrowth advocates might challenge the very notion that ‘true cost’ can express ecological value at all. Finally, conventional perspectives through which technology and innovation are explored in food systems transformation are largely at odds with the promotion of convivial technology (box 18 on the left) in degrowth. However, there is a growing area of food systems literature that offers a critical perspective. This includes responsible innovation and mission-oriented innovation literatures, which could see mutual value in integrating a degrowth lens.

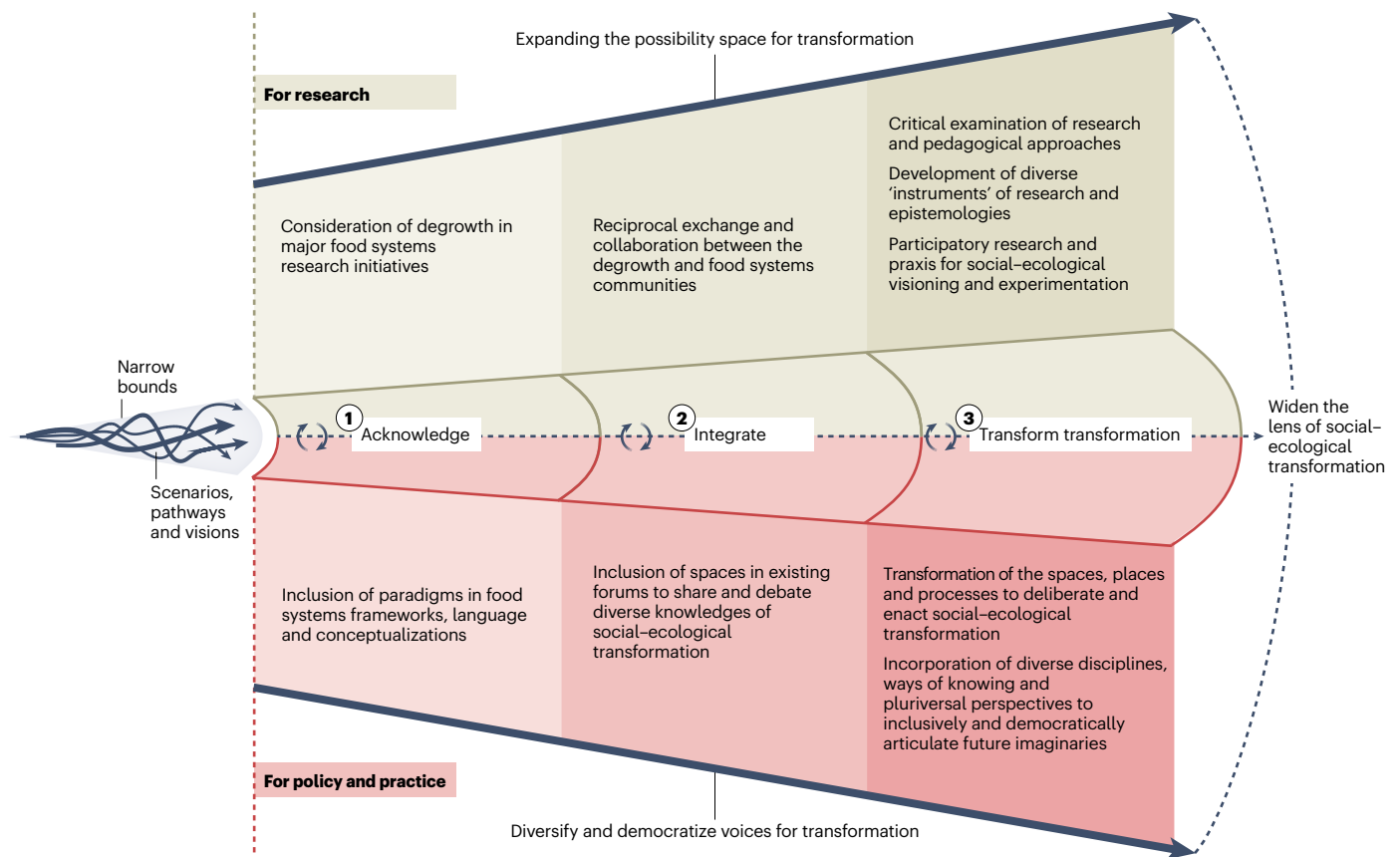


Fig. 2 | Actions to widen the lens of social-ecological transformation. Three levels of action—(1) acknowledge, (2) integrate and (3) transform transformation—for degrowth and food systems research, policy and practice to expand the possibility space and to diversify and democratize voices for transformation.

narratives in global food systems reports¹⁷. Specifically, we systematically identified and interpreted every occurrence of words and phrases with the root word 'growth' (to capture instances of both degrowth and post-growth).

We found no instances across the 52 publications that explicitly mentioned degrowth or post-growth. The closest explicit occurrence was a citation to degrowth research made in the Global Alliance for the Future of Food report *The Politics of Knowledge*¹⁸. As a follow-on analysis, we assessed the reports' treatment of economic growth, clearly noting that the economic aspect of degrowth is but one dimension. Of the 52 publications, 16 were found to include critiques of the general assumption that economic growth is an undisputed desirable goal and is necessary for human well-being and development. The clearest critiques of economic growth came from the International Planning Committee for Food Sovereignty, IPES-Food and the Global Alliance for the Future of Food. We highlight in the next section that degrowth and food systems transformation proposals are, in general, potentially well aligned. This absence therefore risks missing an opportunity to constructively engage with complementary scholarship, narrowing food systems research and action when radical transformation is needed.

Alignment between food systems transformation and degrowth

Figure 1 illustrates the potential alignment of 20 degrowth measures with 20 key food systems transformation measures. This is not intended to offer a comprehensive and exhaustive assessment; rather, it serves to introduce some common degrowth ideas and highlight how these may (or may not) align with key aspects of food systems transformation.

The set of degrowth measures were sourced from a recent review of degrowth policies¹⁹, which identified 50 goals and 100 objectives across 13 themes. We used two criteria to select the 20 measures: (1) the number of references in the reviewed literature and (2) coverage of the overall themes to capture the breadth of policy proposals. Three themes—science and technology, trade, and tourism—do not feature highly by way of referencing literature. To include the measure within these themes, we combined similar measures, which resulted in selecting 17 of the top 20 most frequent proposals and all 13 themes. Altogether, the selected measures cover over half of the total literature references. Food systems measures were sourced from recent literature and global food systems reports^{6,16,20}.

Food system transformation is well aligned, in principle, with common degrowth policies and proposals. For example, the universal provision of fundamental human needs and cultures of sufficiency (boxes 6 and 10 on the left in Fig. 1) translate strongly to health and nutrition goals and demand-side measures for lowering consumption. Indeed, research is ongoing to assess the extent to which food systems can transform to feed ten billion people healthy, sufficient and just diets within planetary boundaries^{8,9}. These efforts represent fertile ground for degrowth perspectives, with questions such as, what are the embedded energy, material and economic costs of such a food systems transformation? What are the possible configurations of production, organization and distribution? And how can efforts to curtail global meat and ultra-processed food intake succeed in a socio-economic system predicated on high levels of consumption and profit maximization?

Another area of interaction, although with uncertainty on the alignment, is the degrowth objective of reducing time in wage labour

Table 1 | Example actions to widen the narrative and analytical lens of social–ecological transformation

Actions	Research	Policy and practice
Acknowledge	Consideration of degrowth in major food systems research initiatives and the need for a more systemic perspective in degrowth approaches concerned with food. Example: EAT–Lancet 2.0 (ref. 9) should be clear about its conceptualization of transformation and, following the lead of the IPCC’s AR6 (ref. 42), include explicit reference to degrowth.	Recognition and inclusion of paradigms in food systems frameworks, language and conceptualizations. Example: Future High Level Panel of Experts reports should amend their food systems definition: “a food system gathers all the elements (environment, people, inputs, processes, infrastructures, institutions, paradigms etc.) and activities...including socio-economic and environmental outcomes”⁴³.
Integrate	Reciprocal exchange and collaboration between the degrowth and food systems transformation research communities. Example: Recent European Union grant funding to degrowth-aligned and post-growth-aligned research, along with the international degrowth and ecological economics conferences, offers forums with which to cross-fertilize food systems and transformation perspectives.	Inclusion of spaces in existing forums to share and debate diverse knowledges of social–ecological transformation. Example: The United Nations Secretary General has published a series of policy briefs focused on ‘Our Common Agenda’ to guide discussions during the United Nations Summit of the Future 2024. This includes a brief on moving beyond GDP and towards “valuing what counts”. This signals a possible entryway for deeper and more diverse debate on growth at the United Nations level, perhaps modelled on the European Parliament’s multistakeholder Beyond Growth 2023 conference.
Transform transformation	Knowledges: critical examination of research and pedagogical approaches. Methods: development of diverse ‘instruments’ of research and epistemologies. Voices: participatory collaboratives of transformative research and praxis for social–ecological visioning and experimentation. Examples: Knowledges: transform economics knowledge structures—for example, a new “Econ 101”⁴⁴ and food systems pedagogies⁴⁵. Methods: evaluate the predominance of neoclassical integrated assessment models and macroeconomic equilibrium models in national and global science–policy interfaces⁴⁶. Learn from studies that have started to push the envelope of such macroeconomic models in other domains⁴⁷ (for example, energy⁴⁸) and build on initial efforts within food systems⁴⁹. Methods: embrace a diversity of methods and approaches—for example, hybrid models, systems dynamics, agent-based models, participatory modelling and non-model critical social science approaches. Voices: diversify and deepen participation while being attentive to power and process in transdisciplinary research⁵⁰.	Transforming the spaces, places and processes to deliberate and enact social–ecological transformation. Incorporating diverse disciplines, ways of knowing and pluriversal perspectives to inclusively and democratically articulate future imaginaries for the planet, including its human and more-than-human worlds. Examples: Learn from the challenges of existing global efforts. IPCC and IPBES science-led panels eschew politics for policy relevance, not policy prescription; COPs suffer from vested-interest actors exploiting the core decision-making process; the Emirates Declaration at COP28, while a good step for finally including food systems, omitted language on dietary shift and overconsumption; and the UNFSS attempted to broaden participation but was critiqued for corporate capture and the creep of multistakeholderism¹¹. Learn from, engage with and platform alternative efforts—for example, the Nyéléni Global Forum of the international food sovereignty movement, the Global Tapestry of Alternatives and the Doughnut Economics Action Lab.

Examples are given for the three action areas (acknowledge, integrate and transform transformation) across two domains (research, and policy and practice). COP, Conference of the Parties; IPCC, Intergovernmental Panel on Climate Change; IPBES, Intergovernmental Science–Policy Platform on Biodiversity and Ecosystem Services.

(box 1 on the left in Fig. 1)¹⁹. For degrowth scholars, reducing time in wage labour is a prerequisite for decreasing production and consumption, improving quality of life and shifting to sustainable practices beyond capitalism²¹. Radical changes to labour and employment pose substantial questions for food systems transformation. For example, from a social–metabolism perspective²², how does a decrease in waged work time affect energetic and nutritional requirements for different populations? Perhaps more significantly, how would the whole-scale shift to agroecological or more labour-intensive production²³ affect labour requirements and the organization of food systems?

Some potential areas of conflict shown in Fig. 1 (red lines—potentially unaligned) are arguments against limiting long-distance trade from a food systems perspective. These include the possibility, in a globalized market-dominated food system, of less economically efficient allocation of production²⁴ and reduced capacity for needs-based allocation of nutrient supply²⁵, although whether the latter is observed is debated²⁶. Furthermore, with around a quarter of food for human consumption being traded²⁷, a future degrowth-aligned Global North with reduced consumption and food imports has the potential, in the absence of other measures, for negative short-term effects on export-oriented economies²⁸.

Another area of possible tension is the promotion of convivial—appropriate, accessible and human-autonomy-centred—technologies among degrowth advocates²⁹. Technological change and innovation have led to productivity increases over the past 100 years that have

enabled food supply to more than keep pace with a growing global population³⁰. Technology and innovation remain a dominant feature of food systems transformation research and policy^{20,30}, but debate and different visions of their role exist. These include visions oriented towards techno-optimism, which bring a technical engineering perspective that ‘technology will save us’ and contrast with a techno-realist view that recognizes that technology both solves and creates problems and may instead emphasize social and structural changes to demand³¹. More recently, literature has emerged on responsible innovation and mission-oriented food systems³². Here the focus is on technology and innovation that serves clear social goals and needs in addition to delivering improvements in productivity or efficiency. A key avenue of future research is to explicitly consider the compatibility of food systems technology and innovation with degrowth approaches (see, for example, the digitalization in food systems³³). Overall, there is a need to continue building the evidence on proposals for degrowth-oriented food systems¹⁹, including new scenarios that explore the implications of post-growth and degrowth-oriented futures for food systems supply, demand and overall structure³⁴, mirroring recent calls for similar scenarios in climate mitigation³⁵.

The challenge of the ‘how’ of transformation

Degrowth scholarship to date has focused on justifying and articulating its objectives, with less research devoted to how a degrowth-oriented sustainability transformation may occur in reality^{19,35} (although this is

beginning to change with works such as *Degrowth & Strategy*³⁶). Similarly, food systems research has focused on the ‘what’, but gaps remain in the ‘how’ of transformation^{20,37}. Indeed, research outputs “rarely shed light on how to implement the desired changes”²⁰. For example, we know that dietary change, including meat reduction, is critical for healthy diets and sustainable food systems, but achieving this at scale means navigating complex and interacting influences. The effectiveness of behavioural interventions is far from settled³⁸, while structural determinants of dietary change (for example, political will, corporate power and socio-cultural norms) remain largely unaddressed.

That the ‘how’ remains elusive is no great mystery. It is easier to assess and communicate what needs to be done than to engage with the difficult work of unpacking means and ends, complex interactions, time dependencies, lock-ins and local contexts of social–ecological transformation. Underlying all this is a diversity of what transformation means and the asymmetric capacity of powerful actors to shape the direction of transformation³⁹. This common challenge should be viewed as a reciprocal invitation for researchers and activists from both communities to critically engage with one another to share priorities, insights, barriers and opportunities. How we conceptualize, approach and enact transformation shapes the kinds of problems we are attentive to, the solutions we propose and the winners and losers of any desired pathway. Both degrowth and food systems transformation could narrow the challenge of ‘how’ by incorporating broader perspectives of sustainability transformation⁴⁰.

Actions to integrate degrowth and food systems transformation perspectives

The integration of degrowth as a plausible pathway for food systems transformation raises important questions. For example, just how compatible is degrowth with predominant visions of food systems transformation? How would a degrowth-oriented food systems transformation be financed? What would be the role of technology, innovation, markets and trade? How might labour, employment and livelihoods be reimaged? And how might food systems governance, organization and ownership be reconfigured in a degrowth-oriented transformation? Research communities should build on recent efforts in the critical social sciences⁴¹ to more broadly develop and articulate a degrowth–food systems research agenda that cuts across diverse scholarly disciplines.

Here we offer a set of practical actions for researchers, policymakers and practitioners to better integrate degrowth and food systems transformation perspectives. We give a conceptual illustration of these actions in Fig. 2 and provide concrete supporting examples in Table 1. The three action areas can be summarized as follows: (1) acknowledge—explicit consideration and recognition of political economic paradigms within food systems transformation research and policy activities (for example, in EAT–Lancet 2.0); (2) integrate—integration of heterodox alternatives such as degrowth for a deeper and more diverse understanding of social–ecological transformation (for example, in the UN Summit of the Future 2024); (3) transform transformation—critical examination and democratization of how we conceptualize, approach and enact transformation (for example, the Global Tapestry of Alternatives). Taken together, these actions are intended to help deconstruct key barriers to transformation that reinforce an unjust, unhealthy, unsustainable status quo.

Growth is the de facto goal of governments and globalized agri-food market forces—growth in trade, growth in production and growth in consumption. The primacy and centrality of growth in our current food and political-economic systems pose enormous challenges to cultivating and nurturing alternatives. To avoid locking in the dominant pathways of change shaped by powerful actors with vested interests, it is critical that food systems transformation engages with radical futures such as degrowth. This engagement needs to go beyond simply mutual acknowledgement, to an integration of degrowth and food systems research

in plausible narratives of change. In doing so, it needs to deepen and diversify how we theorize and enact transformation towards sustainability⁴¹. The global polycrisis necessitates a widening of the narrative and the analytical lens of transformation to explore a wide range of possible futures, beyond those simply deemed more likely.

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Author contributions

M.G., D.M.-D. and M.H. conceptualized the Perspective. M.G. wrote the original draft with input from D.M.-D. The analysis was performed by A.N., M.B.A., M.G. and C.C. All authors provided feedback and edits and contributed to revising the final version of the paper.

Competing interests

The authors declare no competing interests.

Additional information

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